Trust for multi-source RAG under attack: results across datasets and seeds

Generated from the saved runs

Built 2026-10-24 15:49

Scope. 873 analyses over 9 dataset–seed runs (datasets: HotpotQA, NQ, TriviaQA; seeds: 0, 1, 2). Every method answers the same cached LLM answers (ollama_llama3_8b (llama3:8b-instruct-fp16), thesis code at 5b5cee6, 6039b30, not recorded), so differences come only from how sources are trusted and how the answer is chosen. Each setup has 9 sources (5 for the honest mix, 7 for the domain experts) with 200 warm-up and 600 test questions; RA-RAG's answering asks $\kappa = 4$ sources; accuracy uses the fair match.

1 Summary of results

Each result of the first evaluation series is a comparison between a baseline and a method on the same test questions. Table 1 gives the difference on every dataset and over all runs, Figure 1 the same with its interval.

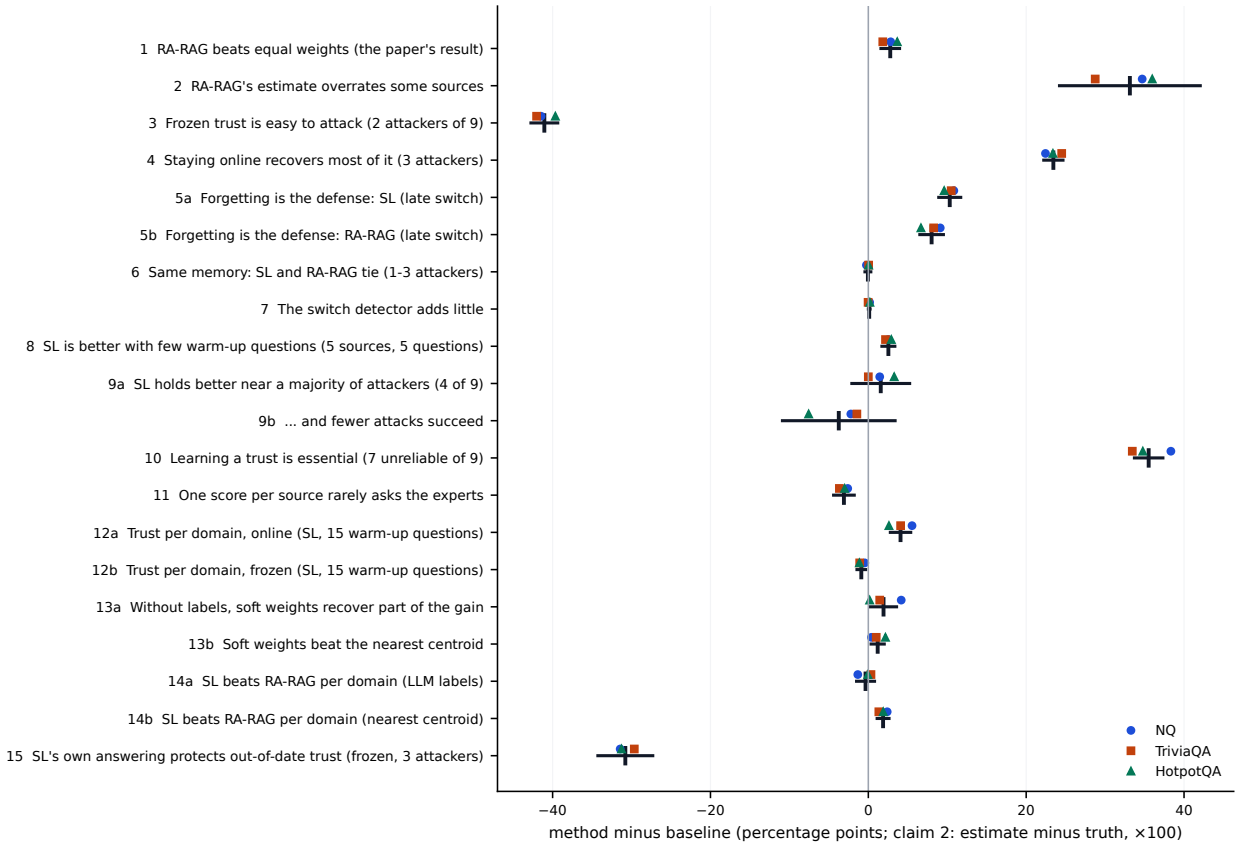


Figure 1: Method minus baseline for each result (percentage points): per dataset (markers) and over every dataset and seed (bar: mean; line: 95% interval).

#	Result (method vs baseline)	NQ	TriviaQA	HotpotQA	All, 95% CI	Holds
1	RA-RAG beats equal weights (the paper's result): RA-RAG (paper) vs equal weights, accuracy	+2.8 (3/3)	+1.8 (2/3)	+3.7 (3/3)	+2.8 [+1.6, +4.0]	8/9
2	RA-RAG's estimate overrates some sources: RA-RAG's estimate, largest excess vs warm-up truth, estimate minus truth (score)	+34.7 (3/3)	+28.7 (3/3)	+36.0 (3/3)	+33.1 [+24.2, +42.1]	9/9
3	Frozen trust is easy to attack (2 attackers of 9): SL with aging vs RA-RAG frozen (paper), attacks that succeed	−41.5 (3/3)	−42.0 (3/3)	−39.7 (3/3)	−41.0 [−42.7, −39.4]	9/9
4	Staying online recovers most of it (3 attackers): RA-RAG online vs RA-RAG frozen, accuracy	+22.4 (3/3)	+24.5 (3/3)	+23.4 (3/3)	+23.4 [+22.2, +24.7]	9/9
5a	Forgetting is the defense: SL (late switch): SL with aging vs SL online, accuracy	+10.8 (3/3)	+10.5 (3/3)	+9.6 (3/3)	+10.3 [+8.9, +11.7]	9/9
5b	Forgetting is the defense: RA-RAG (late switch): RA-RAG with a window vs RA-RAG online, accuracy	+9.1 (3/3)	+8.3 (3/3)	+6.7 (3/3)	+8.0 [+6.5, +9.5]	9/9
6	Same memory: SL and RA-RAG tie (1-3 attackers): SL with a window vs RA-RAG with a window, accuracy	−0.2 (8/9)	+0.0 (7/9)	+0.0 (4/9)	−0.1 [−0.4, +0.3]	19/27
7	The switch detector adds little: SL with aging + the detector vs SL with aging, accuracy	+0.2 (6/6)	−0.0 (6/6)	+0.2 (6/6)	+0.1 [+0.0, +0.2]	18/18
8	SL is better with few warm-up questions (5 sources, 5 questions): SL frozen vs RA-RAG frozen, accuracy	+2.5 (3/3)	+2.2 (3/3)	+2.9 (3/3)	+2.5 [+1.7, +3.4]	9/9
9a	SL holds better near a majority of attackers (4 of 9): SL with a window vs RA-RAG with a window, accuracy	+1.4 (2/3)	+0.0 (1/3)	+3.3 (3/3)	+1.6 [−2.1, +5.2]	6/9
9b	... and fewer attacks succeed: SL with a window vs RA-RAG with a window, attacks that succeed	−2.2 (2/3)	−1.5 (1/3)	−7.6 (3/3)	−3.7 [−10.9, +3.4]	6/9
10	Learning a trust is essential (7 unreliable of 9): best learned trust vs equal weights, accuracy	+38.3 (3/3)	+33.4 (3/3)	+34.8 (3/3)	+35.5 [+33.7, +37.3]	9/9
11	One score per source rarely asks the experts: RA-RAG (paper) vs equal weights, accuracy	−2.6 (2/3)	−3.7 (3/3)	−3.0 (3/3)	−3.1 [−4.4, −1.8]	8/9
12a	Trust per domain, online (SL, 15 warm-up questions): SL per domain vs SL, one score, accuracy	+5.5 (3/3)	+4.1 (3/3)	+2.6 (3/3)	+4.1 [+2.8, +5.4]	9/9
12b	Trust per domain, frozen (SL, 15 warm-up questions): SL frozen per domain vs SL frozen, one score, accuracy	−0.4 (2/3)	−1.1 (1/3)	−1.1 (1/3)	−0.9 [−1.4, −0.3]	4/9
13a	Without labels, soft weights recover part of the gain: SL, soft weights vs SL, no domains, accuracy	+4.2 (3/3)	+1.4 (2/3)	+0.2 (1/3)	+1.9 [+0.3, +3.6]	6/9
13b	Soft weights beat the nearest centroid: SL, soft weights vs SL, nearest centroid, accuracy	+0.4 (2/3)	+1.0 (3/3)	+2.2 (3/3)	+1.2 [+0.4, +2.0]	8/9
14a	SL beats RA-RAG per domain (LLM labels): SL per domain vs RA-RAG per domain, online, accuracy	−1.3 (1/3)	+0.3 (1/3)	−0.1 (2/3)	−0.4 [−1.5, +0.8]	4/9
14b	SL beats RA-RAG per domain (nearest centroid): SL per domain vs RA-RAG per domain, online, accuracy	+2.4 (3/3)	+1.3 (3/3)	+1.9 (3/3)	+1.9 [+1.1, +2.6]	9/9
15	SL's own answering protects out-of-date trust (frozen, 3 attackers): SL's own answering vs RA-RAG's answering, attacks that succeed	−31.4 (3/3)	−29.7 (3/3)	−31.3 (3/3)	−30.8 [−34.3, −27.3]	9/9
16a	Fair test: trust per domain from an independent labeller's labels: SL per domain, labeller B vs SL, one score, accuracy	–	–	–	–	–
16b	Fair test: soft weights against the labeller's own labels: SL, soft weights from B vs SL, labeller B's labels, accuracy	–	–	–	–	–
16c	The cost of label noise: the experts' own labels against labeller B's: SL, the experts' labels vs SL, labeller B, accuracy	–	–	–	–	–
16d	Fair test: SL beats RA-RAG per domain (labeller B): SL per domain vs RA-RAG per domain, online, accuracy	–	–	–	–	–
17a	Thin attacks (10% of questions): aging still beats frozen trust: SL with aging vs RA-RAG frozen, attacks that succeed	–	–	–	–	–
17b	Attacks in bursts (20%): aging still beats frozen trust: SL with aging vs RA-RAG frozen, attacks that succeed	–	–	–	–	–
17c	Thin attacks: SL's own answering protects out-of-date trust: SL's own answering vs RA-RAG's answering, attacks that succeed	–	–	–	–	–

Table 1: The first evaluation series' results across datasets and seeds. Each cell is the method's difference from the baseline in percentage points (accuracy or attacks that succeed), averaged over seeds, and in brackets how many of that dataset's seeds go the expected way; "All" averages every dataset and seed, with a 95% t interval when there are several; "Holds" counts the runs, over every dataset and seed, where the difference goes the expected way (for a tie: within one point).

2 Attacks and forgetting

Coordinated attackers answer honestly through the warm-up, then all give the same wrong answer on half the test questions. A trust frozen after the warm-up keeps trusting them; a trust that keeps learning and forgets old evidence does not. Tables 2 and 3, Figure 2.

Trust	1 of 9	2 of 9	3 of 9	4 of 9	switch
NQ					
RA-RAG, frozen (paper)	84.5 $\pm$0.6	69.0 $\pm$0.7	58.1 $\pm$1.2	52.1 $\pm$0.3	73.4 $\pm$1.0
RA-RAG online	86.2 $\pm$0.3	84.3 $\pm$1.7	80.5 $\pm$0.9	51.8 $\pm$0.4	76.9 $\pm$1.9
RA-RAG, window of 100	85.8 $\pm$1.0	85.4 $\pm$0.6	83.8 $\pm$0.4	59.3 $\pm$2.8	86.0 $\pm$0.6
SL, frozen	84.8 $\pm$1.3	69.2 $\pm$0.3	58.4 $\pm$1.2	52.4 $\pm$0.7	73.7 $\pm$1.2
SL online, no forgetting	85.7 $\pm$1.0	83.8 $\pm$1.6	78.8 $\pm$1.2	52.6 $\pm$0.5	74.0 $\pm$1.5
SL, aging $\lambda = 0.97$	85.7 $\pm$1.0	85.4 $\pm$0.5	83.4 $\pm$0.8	60.0 $\pm$3.2	84.8 $\pm$1.3
SL, window of 100	85.8 $\pm$1.1	85.2 $\pm$0.9	83.3 $\pm$0.7	60.8 $\pm$3.2	84.5 $\pm$1.0
Equal weights	85.8 $\pm$0.2	81.6 $\pm$1.1	75.3 $\pm$0.3	55.8 $\pm$0.9	81.0 $\pm$0.6
TriviaQA					
RA-RAG, frozen (paper)	90.1 $\pm$0.3	73.4 $\pm$1.5	61.3 $\pm$0.2	56.4 $\pm$0.7	78.9 $\pm$1.5
RA-RAG online	91.9 $\pm$0.8	91.0 $\pm$0.9	85.8 $\pm$1.0	56.2 $\pm$0.9	83.4 $\pm$3.0
RA-RAG, window of 100	91.9 $\pm$0.8	91.7 $\pm$0.6	89.7 $\pm$0.8	67.2 $\pm$5.9	91.7 $\pm$1.0
SL, frozen	90.3 $\pm$0.9	73.1 $\pm$1.2	63.2 $\pm$1.5	56.7 $\pm$0.3	78.7 $\pm$1.3
SL online, no forgetting	92.2 $\pm$1.0	90.0 $\pm$1.2	85.0 $\pm$0.6	57.1 $\pm$1.2	81.2 $\pm$3.5
SL, aging $\lambda = 0.97$	91.8 $\pm$0.8	91.8 $\pm$1.0	89.9 $\pm$1.1	65.6 $\pm$4.1	91.7 $\pm$0.8
SL, window of 100	92.2 $\pm$0.6	91.8 $\pm$0.6	89.5 $\pm$0.6	67.2 $\pm$3.8	91.6 $\pm$1.3
Equal weights	91.7 $\pm$0.5	87.8 $\pm$0.3	83.6 $\pm$1.0	64.3 $\pm$1.3	87.9 $\pm$0.5
HotpotQA					
RA-RAG, frozen (paper)	77.2 $\pm$1.9	62.2 $\pm$1.9	52.1 $\pm$1.1	47.1 $\pm$0.3	67.1 $\pm$1.8
RA-RAG online	79.6 $\pm$0.8	77.9 $\pm$1.3	75.5 $\pm$2.1	47.2 $\pm$0.2	71.5 $\pm$2.5
RA-RAG, window of 100	79.5 $\pm$0.9	78.8 $\pm$1.2	77.3 $\pm$0.8	52.7 $\pm$2.8	78.2 $\pm$1.4
SL, frozen	77.7 $\pm$2.0	62.6 $\pm$2.6	52.1 $\pm$1.3	47.1 $\pm$0.4	67.1 $\pm$1.9
SL online, no forgetting	79.7 $\pm$1.0	77.8 $\pm$1.9	72.3 $\pm$3.8	47.8 $\pm$0.3	67.5 $\pm$2.3
SL, aging $\lambda = 0.97$	80.1 $\pm$0.7	79.8 $\pm$1.3	76.6 $\pm$1.3	55.8 $\pm$4.8	77.1 $\pm$1.8
SL, window of 100	80.0 $\pm$0.7	79.5 $\pm$0.9	76.2 $\pm$1.8	55.9 $\pm$5.2	77.0 $\pm$2.0
Equal weights	80.3 $\pm$1.0	75.9 $\pm$2.1	70.3 $\pm$1.4	51.5 $\pm$1.0	75.6 $\pm$1.2

Table 2: Coordinated build-then-betray attackers among 9 sources (and 3 attackers switching halfway through the test): accuracy in percent, mean over seeds with the spread as a subscript.

Trust	1 of 9	2 of 9	3 of 9	4 of 9	switch
NQ					
RA-RAG, frozen (paper)	9.4 ±1.2	43.8 ±2.0	64.0 ±3.9	73.2 ±2.2	57.5 ±4.6
RA-RAG online	1.7 ±1.1	4.8 ±2.6	12.5 ±1.4	73.9 ±1.6	43.4 ±9.1
RA-RAG, window of 100	0.9 ±0.7	1.9 ±1.3	5.8 ±0.9	58.3 ±4.5	6.7 ±2.9
SL, frozen	7.0 ±4.9	43.5 ±2.1	63.0 ±3.7	72.8 ±3.0	57.3 ±6.4
SL online, no forgetting	1.3 ±0.6	4.9 ±1.9	16.8 ±1.9	72.8 ±3.3	54.8 ±7.5
SL, aging $\lambda = 0.97$	0.7 ±0.3	2.3 ±1.1	7.3 ±1.9	55.9 ±9.2	9.3 ±3.4
SL, window of 100	0.7 ±0.3	2.2 ±1.0	7.4 ±0.8	56.1 ±8.2	9.9 ±5.5
Equal weights	1.7 ±1.0	12.8 ±0.9	25.6 ±3.6	65.4 ±3.7	20.4 ±3.2
TriviaQA					
RA-RAG, frozen (paper)	6.5 ±0.8	44.0 ±2.6	64.5 ±1.7	73.4 ±1.9	59.8 ±6.7
RA-RAG online	0.6 ±0.6	4.3 ±0.8	12.4 ±2.8	73.4 ±1.8	39.9 ±13.1
RA-RAG, window of 100	0.9 ±0.9	2.2 ±0.5	6.2 ±2.0	51.8 ±10.8	5.6 ±3.5
SL, frozen	7.4 ±1.5	43.7 ±2.0	61.3 ±3.7	73.0 ±1.5	60.7 ±6.4
SL online, no forgetting	0.6 ±0.6	5.3 ±0.4	15.0 ±1.7	71.6 ±2.4	50.8 ±13.6
SL, aging $\lambda = 0.97$	0.9 ±0.9	1.9 ±0.5	5.1 ±0.6	53.2 ±8.6	5.1 ±2.4
SL, window of 100	0.9 ±0.9	2.2 ±0.3	5.8 ±0.4	50.3 ±8.8	7.3 ±6.1
Equal weights	0.6 ±0.6	9.1 ±1.6	15.9 ±3.1	55.8 ±1.5	13.2 ±5.1
HotpotQA					
RA-RAG, frozen (paper)	10.3 ±3.7	41.7 ±2.0	59.6 ±1.3	67.0 ±1.0	55.8 ±4.8
RA-RAG online	1.3 ±1.7	4.1 ±3.1	10.4 ±5.1	67.3 ±1.0	32.7 ±10.7
RA-RAG, window of 100	0.5 ±0.5	1.6 ±1.0	6.0 ±2.0	56.7 ±5.5	7.0 ±4.0
SL, frozen	10.3 ±3.2	40.4 ±1.6	59.5 ±1.1	67.3 ±0.7	56.5 ±4.8
SL online, no forgetting	1.1 ±1.4	4.4 ±2.1	17.7 ±8.1	67.1 ±1.6	53.3 ±6.7
SL, aging $\lambda = 0.97$	0.7 ±0.8	2.1 ±0.9	7.6 ±2.7	50.9 ±9.6	10.6 ±4.5
SL, window of 100	0.5 ±0.5	2.4 ±0.4	8.4 ±2.8	49.1 ±10.0	11.8 ±6.7
Equal weights	0.9 ±0.6	11.0 ±2.1	22.1 ±2.5	58.6 ±2.4	19.8 ±3.8

Table 3: The same setups: share of attacked questions answered with the attackers' answer, in percent.

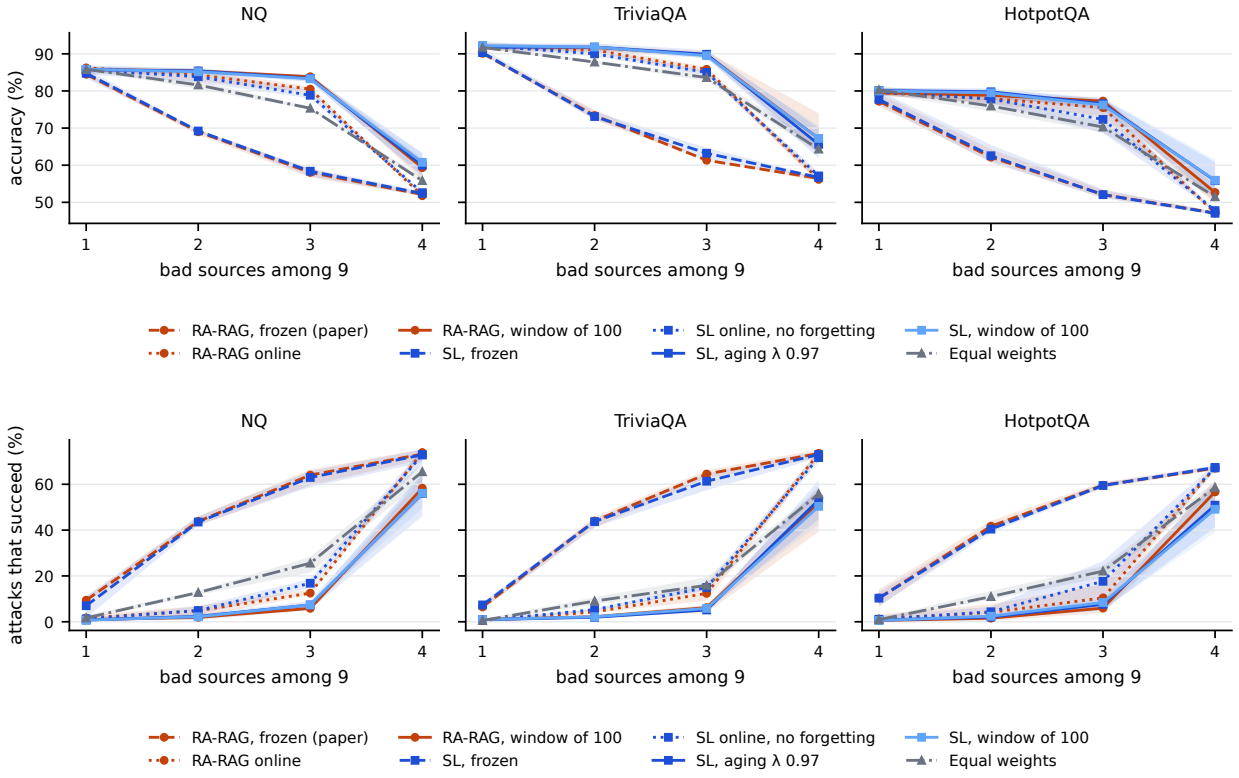


Figure 2: Accuracy (top) and attacks that succeed (bottom) against the number of attackers among 9 sources; mean over seeds, band: range over seeds.

3 Adaptive attackers

Attackers who betray a few questions at a time try to stay under the forgetting rate: 3 of 9 attack only a share of the test questions, spread thinly or in bursts of 5 attacked questions followed by 20 honest ones. The 50% column is the build-then-betray setup above.
No results yet.

4 Unreliable sources

Unreliable sources are right only 10% of the time and disagree with each other. Learning any trust removes most of their harm; equal weights does not (Table 4).

Trust	3 of 9	7 of 9
NQ		
RA-RAG, frozen (paper)	84.7 ±0.6	69.2 ±1.4
RA-RAG online	85.0 ±0.5	69.9 ±2.4
RA-RAG, window of 100	84.8 ±0.9	69.3 ±1.5
SL, frozen	85.1 ±0.3	70.1 ±1.1
SL online, no forgetting	84.3 ±0.4	69.7 ±1.3
SL, aging $\lambda = 0.97$	84.3 ±1.4	69.2 ±1.4
SL, window of 100	84.3 ±1.4	69.2 ±1.5
Equal weights	81.8 ±0.8	32.8 ±1.6
TriviaQA		
RA-RAG, frozen (paper)	90.6 ±1.3	78.1 ±1.8
RA-RAG online	91.0 ±1.0	78.5 ±0.7
RA-RAG, window of 100	91.3 ±0.3	77.6 ±2.0
SL, frozen	91.1 ±0.2	77.6 ±1.8
SL online, no forgetting	91.2 ±0.3	78.8 ±0.6
SL, aging $\lambda = 0.97$	91.6 ±0.5	78.7 ±2.2
SL, window of 100	91.4 ±0.6	77.8 ±1.6
Equal weights	88.8 ±1.0	45.9 ±0.5
HotpotQA		
RA-RAG, frozen (paper)	78.6 ±0.8	63.1 ±1.3
RA-RAG online	78.5 ±0.9	63.1 ±1.1
RA-RAG, window of 100	78.9 ±0.5	63.2 ±1.4
SL, frozen	79.4 ±0.7	63.8 ±1.3
SL online, no forgetting	78.7 ±0.6	63.6 ±1.4
SL, aging $\lambda = 0.97$	78.9 ±0.4	61.7 ±2.6
SL, window of 100	78.8 ±0.5	61.2 ±2.0
Equal weights	74.9 ±0.9	29.4 ±1.0

Table 4: Static unreliable sources (right 10% of the time) among 9: accuracy in percent.

5 Few warm-up questions

With a handful of warm-up answers, RA-RAG's estimate (a raw share of agreements) swings to extremes; SL's trust starts from a prior and moves with the evidence (Table 5, Figure 3).

Dataset	5	10	20
NQ	+2.5	+1.1	−0.3
TriviaQA	+2.2	+0.4	+0.6
HotpotQA	+2.9	+0.2	−0.2

Table 5: Few warm-up questions (five honest sources, reliabilities 0.53–0.91): SL's frozen trust minus RA-RAG's frozen estimate, accuracy in points, by number of warm-up questions; mean over random warm-up draws and seeds.

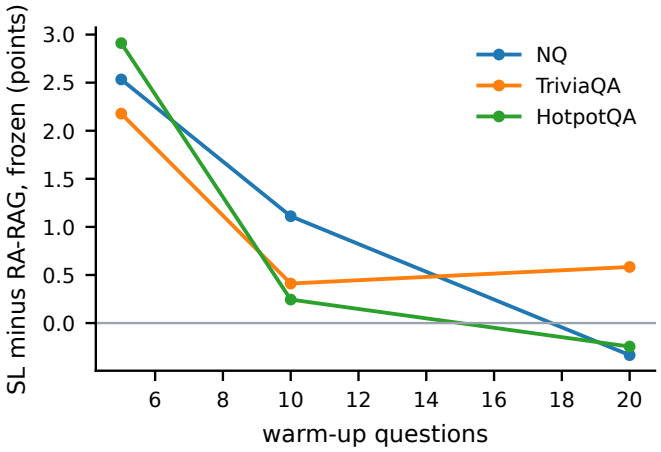


Figure 3: SL's frozen trust minus RA-RAG's frozen estimate (accuracy points) by number of warm-up questions.

6 Domain experts

Each expert is right 90% of the time in its own domain and 30% elsewhere; two generalists are right 60% of the time everywhere. The system is never told who is an expert. Trust is learned per domain, with each question placed among the domains by a label, by the nearest domain centroid of its embedding, or by soft weights over the domains. The experts are built from one LLM's labels (A), so placing questions by those labels is an oracle. The fair test places them by an independent labeller's labels only (B, another model family; centroids from B's warm-up labels too), with A as the upper bound and one score per source as the lower (Table 6, Figure 4).

Domain of a question	NQ		TriviaQA		HotpotQA	
	SL	RA-RAG	SL	RA-RAG	SL	RA-RAG
No domains	58.8 ±0.8	55.3 ±0.6	69.4 ±3.7	67.2 ±2.2	56.1 ±1.3	52.2 ±2.8
Labels A (oracle)	65.0 ±1.4	66.3 ±1.8	75.2 ±0.4	74.8 ±1.9	58.2 ±0.5	58.3 ±1.0
Nearest centroid (A)	62.6 ±0.8	60.2 ±0.4	69.8 ±1.5	68.5 ±1.5	54.1 ±2.4	52.2 ±2.0
Soft weights (A)	62.9 ±0.9	60.2 ±0.4	70.8 ±1.6	68.5 ±1.5	56.2 ±1.8	52.2 ±2.0

Table 6: Domain experts (right 90% of the time in their domain, 30% elsewhere): accuracy with trust learned per domain online, by how each question is placed among the domains, full warm-up. A: the labels the experts are built from, so placing by them is an oracle; B: an independent labeller's, the fair test. RA-RAG's estimator runs per domain on each question's top domain.

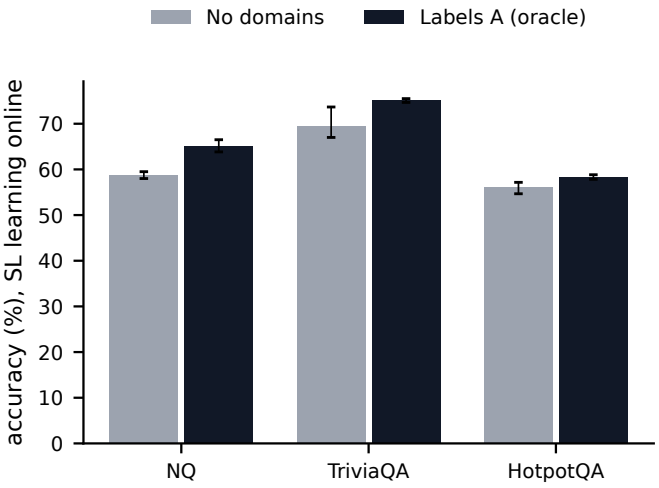


Figure 4: SL's accuracy with trust per domain, by how questions are placed among domains: one score, the oracle labels (A), and the fair test from labeller B (mean over seeds, bar: range).

7 SL's own answering

Instead of RA-RAG's vote over the four most trusted sources, SL can answer by itself: it asks every source, fuses their evidence, and answers, says *unknown*, or says *conflict* and shows two answers (Table 7).

Setup	Answers	Right	RA-RAG, same cov.	Conflict right	Unknown wrong	Attacks succeed SL	RA-RAG
3 unreliable of 9	69.6 ±3.8	90.6 ±4.3	89.5 ±5.0	72.5 ±10.8	40.1 ±6.0	–	–
7 unreliable of 9	0.0 ±0.0	–	–	100.0	49.8 ±2.8	–	–
5 honest sources	44.7 ±4.0	89.7 ±3.3	85.9 ±6.8	72.4 ±5.9	38.2 ±4.4	–	–
1 attacker of 9	86.5 ±2.6	89.3 ±3.6	90.0 ±4.7	75.9 ±12.9	42.3 ±8.2	0.1 ±0.2	0.8 ±0.6
2 attackers of 9	84.6 ±2.6	89.1 ±3.9	89.9 ±4.7	69.6 ±7.3	44.4 ±7.5	2.4 ±1.1	2.1 ±0.7
3 attackers of 9	81.0 ±2.1	84.7 ±3.9	88.7 ±5.1	82.4 ±6.4	43.5 ±8.2	11.4 ±2.5	6.7 ±2.0
4 attackers of 9	77.3 ±1.6	69.1 ±3.0	67.6 ±5.0	88.7 ±5.2	38.3 ±9.4	34.9 ±4.5	53.3 ±8.2
3 attackers, late switch	84.6 ±2.7	87.4 ±3.5	89.5 ±5.1	81.7 ±6.3	38.2 ±13.8	10.6 ±3.4	8.3 ±3.9
domain experts	3.5 ±1.6	86.6 ±5.2	–	77.3 ±8.1	45.1 ±3.3	–	–

Table 7: SL's own answering (answer, *unknown* or *conflict*) with its online trust and aging: share of questions answered, accuracy on those, RA-RAG's accuracy on its most confident questions at the same coverage and trust, conflicts whose two answers include the right one, *unknown* questions whose top answer would have been wrong, and attacks that succeed with SL's answering and with RA-RAG's on the same trust. Means over all datasets and seeds.

A Runs

Dataset	seed 0	seed 1	seed 2
NQ	97/138	97/138	97/138
TriviaQA	97/138	97/138	97/138
HotpotQA	97/138	97/138	97/138

Table 8: Analyses saved per dataset and seed, out of those planned.